



2.3 Quadratic functions and the discriminant

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A quadratic $f(x)=ax^2+bx+c$ ($a \neq 0$) has graph a parabola with vertex at $x=-b/(2a)$, found in vertex form $f(x)=a(x-h)^2+k$. The discriminant $\Delta=b^2-4ac$ determines the nature of the roots: $\Delta>0$ gives two distinct real roots, $\Delta=0$ gives one repeated real root (the graph touches the x-axis), and $\Delta<0$ gives no real roots (two complex conjugate roots).

Worked example: follow the reasoning

Find the discriminant of $2x^2-5x+3=0$ and state the nature of its roots, then solve it.
 $\Delta=(-5)^2-4(2)(3)=25-24=1>0$, so two distinct real roots.
 $x=(5 \pm \sqrt{1})/4 = (5 \pm 1)/4$, giving $x=1.5$ or $x=1$.

Quick check · multiple choice

For what value of k does $x^2+6x+k=0$ have exactly one repeated real root?

- A $k=3$
- B $k=6$
- C $k=9$
- D $k=12$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider $f(x) = 3x^2 - 12x + 7$. [8]
- (a) Write $f(x)$ in the form $a(x-h)^2+k$, and hence state the coordinates of the vertex.
 - (b) Find the discriminant of $f(x)=0$ and state the number of real roots.
 - (c) Find the set of values of m for which $mx^2-12x+7=0$ has no real solutions ($m \neq 0$).
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Final answer: _____



Answers and explanation

Question 1 [8]

(a) $3x^2 - 12x + 7 = 3(x^2 - 4x) + 7 = 3(x-2)^2 - 12 + 7 = 3(x-2)^2 - 5$. Vertex (2, -5).

(b) $\Delta = (-12)^2 - 4(3)(7) = 144 - 84 = 60 > 0$, so two distinct real roots.

(c) Need $\Delta < 0$: $144 - 4m(7) < 0 \rightarrow 144 - 28m < 0 \rightarrow m > 144/28 = 36/7$. So $m > 36/7$ (and $m \neq 0$ already satisfied).

Quick check answer

C. Repeated root needs $\Delta = 0$: $36 - 4k = 0$, so $k = 9$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.